

Section A

Answer **all** the questions in this section

1. **Figure 1.1** below shows 2 organelles from an animal cell.

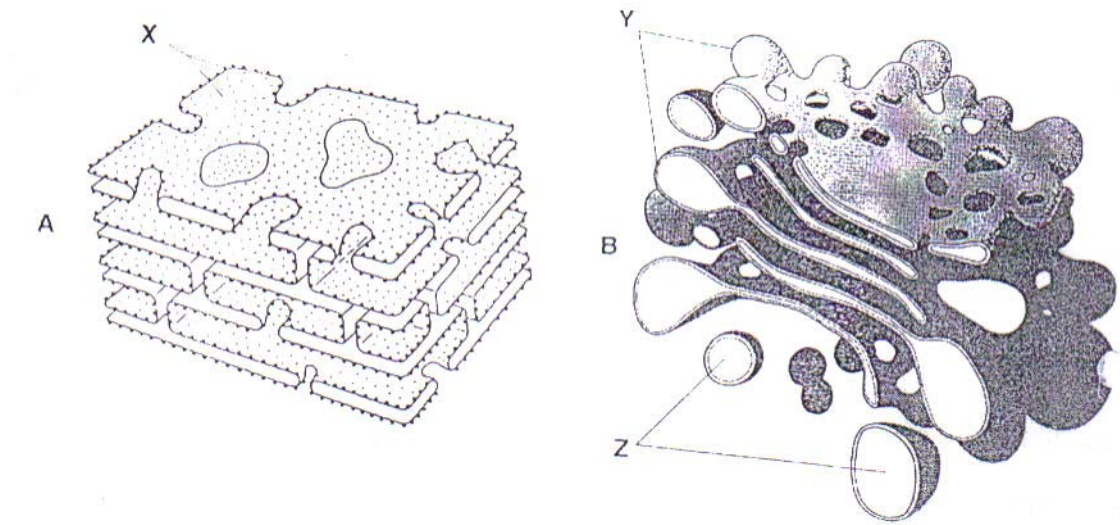


Fig 1.1

- (a) (i) Name the Organelles A and B [1]

A: [Rough Endoplasmic Reticulum](#)

B: [Golgi apparatus](#)

- (ii) Underline the cells in the following list that do not have any A. [1]

liver cell; bacterium; virus; muscle fibre

- (iii) Name the structures Y and Z shown in Fig. B. [1]

Y: [Golgi body / cisternae](#)

Z: [Golgi vesicles/ Secretory vesicles](#)

(iii) Explain the function of A and B with relation to protein synthesis [4]

- Ribosomes on RER – site of protein synthesis ;
- RER transports protein made by ribosomes through the cisternae ;
- Golgi: Chemical modification e.g adding carbohydrate chain ;
- Transport through golgi vesicles/ secretory vesicles to within cell/ out of cell

Figure 1.2 below shows diagrams that represent certain biomolecules found in an animal cell.

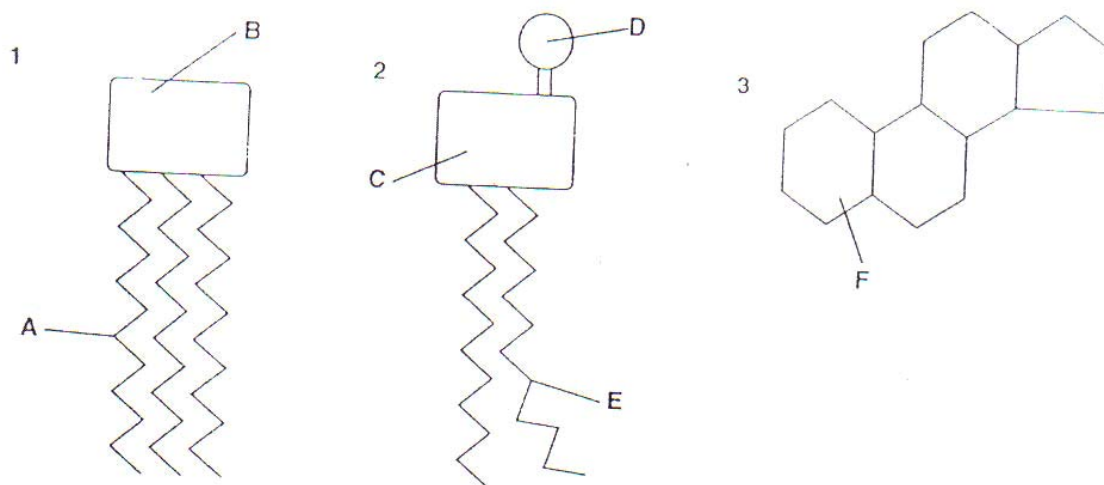


Fig 1.2

(b) (i) Which are the three most common elements in these molecules? [1]

- Carbon, Hydrogen and Oxygen

(ii) Give the number of the molecules (1, 2 or 3) which contains a fourth element and name this additional element. [1]

- 2 – Phosphorus

(c) (i) What general name is given to molecules with a structure similar to diagram 3? [1]

- Steroids

(ii) Suggest one important function in mammals for this type of molecule. [1]

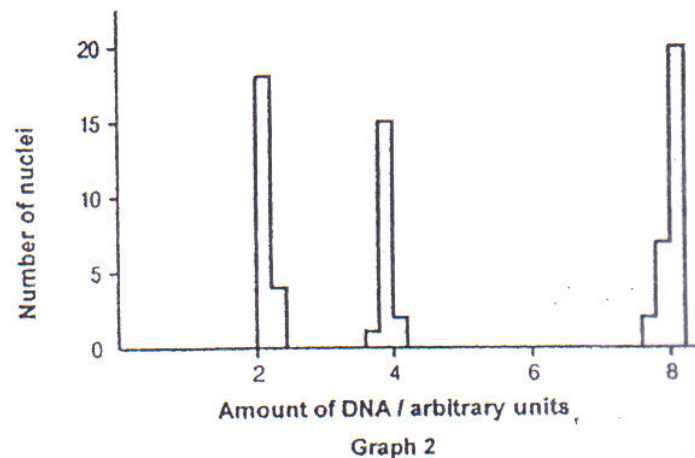
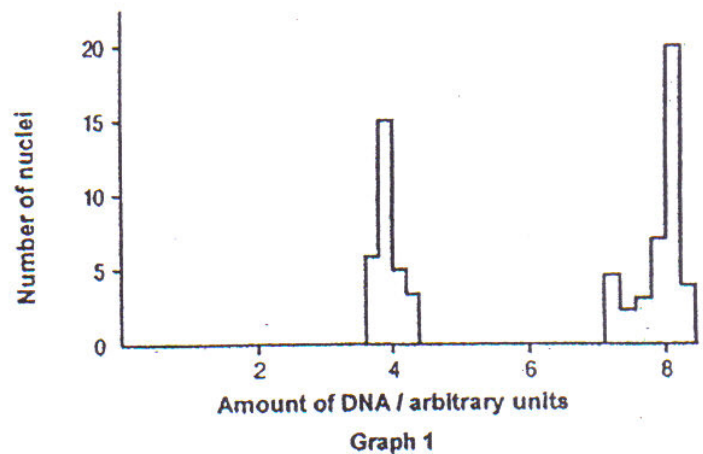
- Cholesterol regulate fluidity of the membrane phospholipids bilayer
- Steroid hormone for regulation (state example)

(iii) Explain how the structure of molecule 2 is related to its function in the cell membranes. [3]

- Phosphate group is charged and is hydrophilic, hydrocarbon tail is hydrophobic OR amphipathic nature of molecule 2.

[Total: 14 marks]

2. The formation of sperm cells involves mitosis and meiosis. The graphs below show the amount of DNA in the nuclei of cells taken from different parts of a mammalian testis.



With reference to the graphs,

(a) (i) Explain the evidence which shows that Graph 1 represents cells undergoing mitosis. [1]

- Two peaks (4 and 8 DNA arbitrary units) representing DNA replication and cytokinesis.
- Absence of a 3rd peak/ group of cells with ~2 DNA arbitrary units.

(ii) Explain how the events of meiosis result in three groups of nuclei in Graph 2. [2]

- Parent cells have ~3.5-4.2 (range) units of DNA in 1st group of cells, cells undergo DNA replication in interphase I
- Homologous chromosomes separate during anaphase I, after telophase II, cells will have half the amount of DNA

(b) Explain how meiosis can produce daughter cells with slightly different genetic makeup. [2]

- Crossing over/ exchange of genetic material between non-sister chromatids of homologous chromosomes.
- Independent assortment of homologous chromosomes at metaphase I/ metaphase II OR Segregation of the homologous chromosomes.

(c) Cancer is a disease caused by altered gene expression. Describe how an abnormal p53 gene can give rise to a cancerous cell. [3]

- Abnormal p53 protein is produced OR abnormal gene does not code for transcription factor.
- Fails to stop cell division, repair DNA and induce apoptosis.
- Cells with damaged DNA proliferate/ continue to divide.

[Total: 8 marks]

3. **Figure 3.1** shows a T4 bacteriophage.

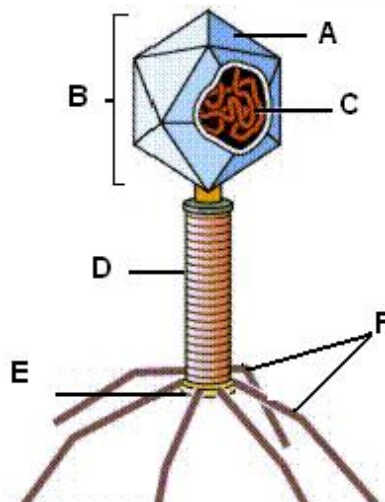


Fig 3.1

(a) (i) Explain what is meant by the term bacteriophage [1]

- Virus that infects bacteria

(ii) Identify the structures labeled A to F [3]

A: Capsid protein

D: Tail sheath

B: Head/ Capsid

E: Tail/end/base plate

C: Deoxyribonucleic acid/ genetic material

F: Tail fibres

(b) The Human Immunodeficiency Virus (HIV) is a retrovirus. Outline the main events of what happens after a HIV virus enters host cell. [4]

- Double strand viral + reverse transcriptase
- Proviral intergration + integrase
- Transcription to
- Synthesis of viral proteins
- Assembly of new virions
- Budding off, acquires host envelope + viral glycoproteins

Figure 3.2 shows a TTP molecule.

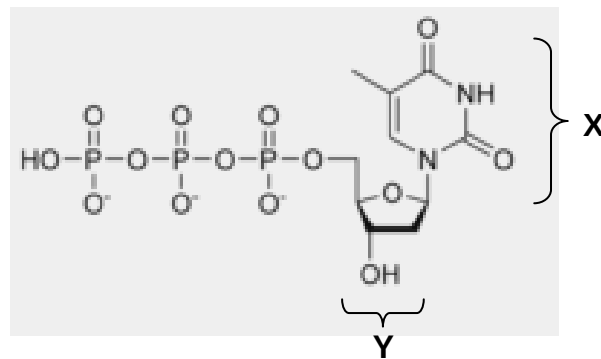


Fig 3.2

(c) (i) Name the compounds labeled X and Y.[1]

X Nitrogenous base/

Y Ribose sugar

Azidothymidine (AZT), a nucleoside analogue of TTP, is used as an anti-retroviral medication to treat AIDS. The normal 3' OH group of the sugar is replaced by an azido functional group. It works as a competitive inhibitor to reverse transcriptase. Incorporation of this analogue results in chain termination.

(d) Suggest a mechanism of how this drug works. [2]

- Absence of free 3' OH group
- AZT structurally similar to TTP
- Competes for the active site of RT

[Total: 11 marks]

4. **Figure 4.1** illustrates the binding of a protein with a DNA molecule.

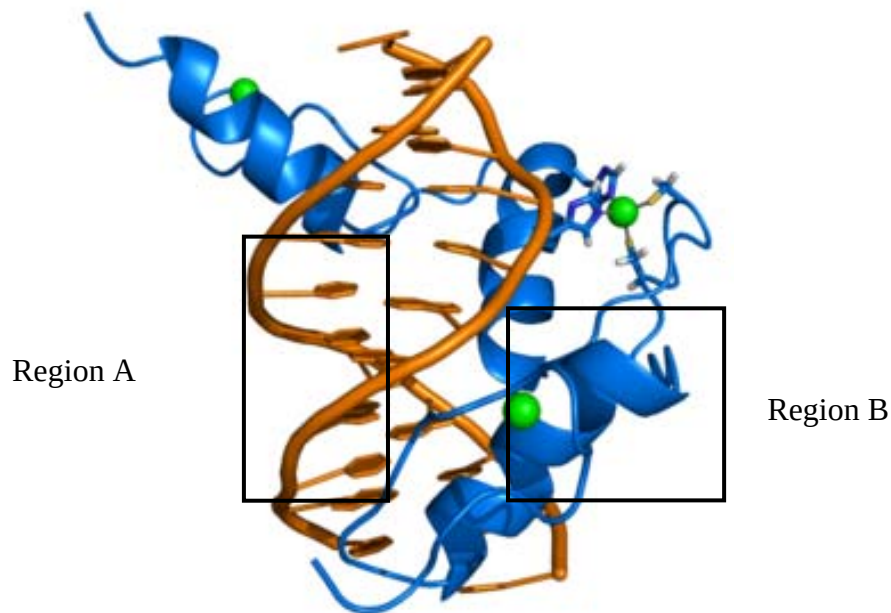


Fig 4.1 Zinc finger DNA complex

(a) With reference to **Fig 4.1**,

(i) Describe in what ways the helical structure shown in Region A and B differ. [3]

- Double strand (A) vs one single polypeptide (B)
- For A each turn is 10 base pair, B is 3.6 amino acids per turn.
- Bonds in A phosphodiester and hydrogen, B has peptide, ionic, hydrogen bonds.

Region B contains a zinc finger protein motif. It is a highly conserved protein motif and the binding specificity is conferred by a short stretch within the α -helix domain. Transcription factors often contain zinc finger motifs.

(ii) Suggest why a transcription factor only binds to certain part of a DNA molecule. [3]

- Transcription factor has binding site
- Complementary
- Different sequences of bases have different conformation/shape

Figure 4.2 shows a type of cell receptor molecule.

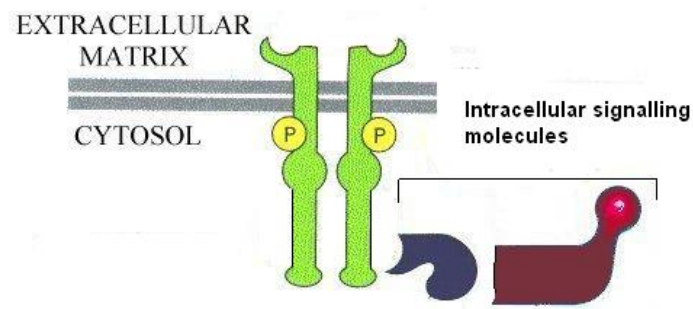


Fig 4.2

(b) (i) State the three main stages of cell signaling. [1]

- reception, transduction, response

(c) With reference to **Fig 4.2**,

(i) State the type of receptor shown in the diagram. [1]

- Tyrosine kinase receptor/ Receptor tyrosine kinase

(ii) Explain how it transmits information from the extracellular to the intracellular environment [2]

- Binding of signal molecules brings two subunits/ dimer close
- Autophosphorylation triggers
- Assembly of an intracellular signaling complex

(d) Ras is a G-protein involved in signal transduction. Ras is also known as a proto-oncogene. When mutated, it often is permanently activated resulting in cancer.

(i) Explain what is signal transduction. [2]

- Relay information
- From signal to get a cellular response
- Conversion by conformational change or change in type/form of molecule
- Phosphorylation cascade by enzyme amplify cell's response to signal.

[Total: 12 marks]

5. **Figure 5.1** shows a diagram of a telomere. Telomeres possess a unique single stranded G-rich 3' overhang that loops back to form a hairpin structure as shown.

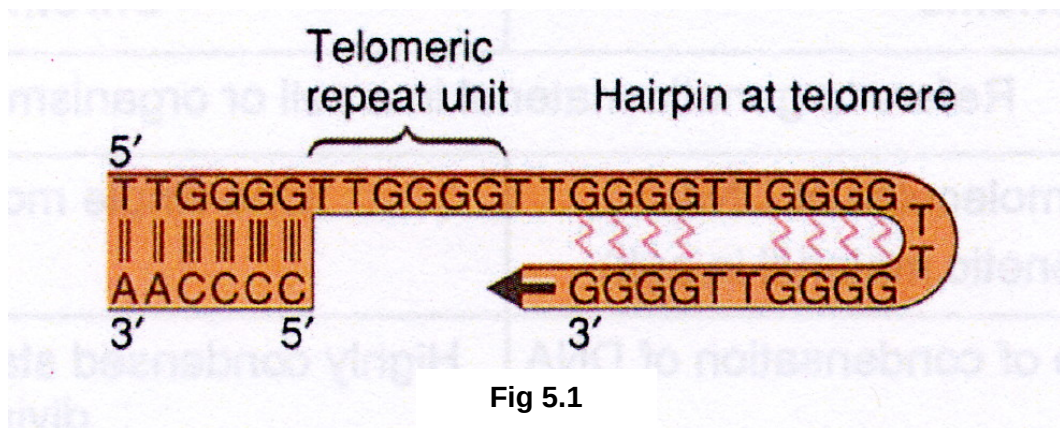
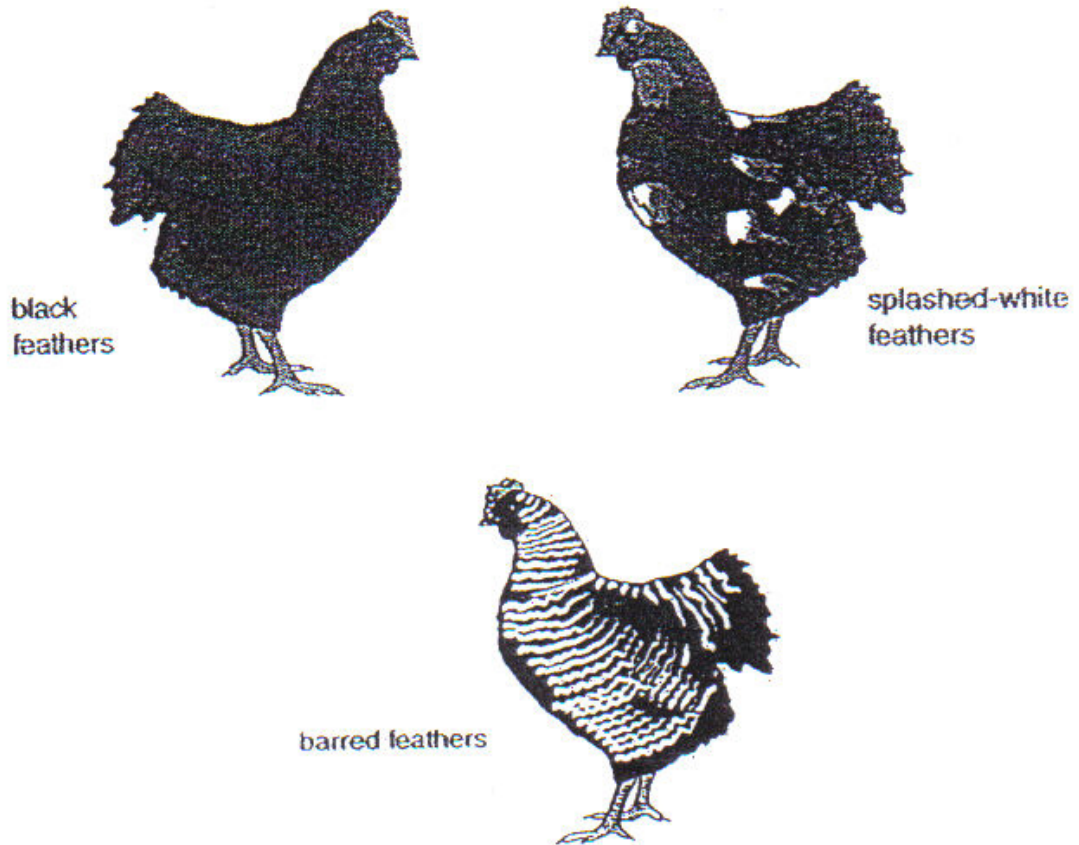


Fig 5.1

- (a) Describe the role of telomeres [3]
- Repetitive sequence found at the end of a chromosome.
 - Plays a protective function because it delays degradation of genes near the end of the telomeres.
 - Prevent ends of chromosomes from fusing which makes a genome unstable
 - Provide counting mechanism for cells to undergo apoptosis when the telomere becomes too short.
- (b) With reference to **Fig 5.1**,
- (i) Describe the term tandem repeats [1]
- Consecutive repeats directly adjacent to each other.
- (c) The telomerase enzyme is responsible for extending the telomere. It is essential in single celled organisms, while in humans and other vertebrates, it is active in reproductive cells.
- (i) Explain how the 3' overhang is formed at the end of a linear chromosome [3]
- Numerous RNA primers formed on lagging strand + replaced with DNA nucleotides
 - Primer of 5' end removed but cannot be replaced because it does not have a free 3' OH group for DNA polymerase to add primers
 - Results in a 3' SS DNA
- (ii) Explain why regeneration of telomeres is not normally required in human body cells. [2]
- Most human cells required to divide a limited number of times
 - After which the cell undergoes apoptosis/programme cell death when the telomeres are too short.

[Total: 9 marks]

6. A gene for feather colour in chickens is carried on an autosome. This gene has two alleles, black (C^B) and splashed-white (C^W). When a male chicken with black feathers is mated with a female chicken with splashed-white feathers, all the offspring have blue feathers. This also occurs when a male chicken with splashed-white feathers is crossed with a female with black feathers.



Another gene may cause stripes on feathers (barred feathers). The gene is carried on the X chromosome. The allele for barred feathers (X^A) is dominant to the allele for non-barred feathers (X^a). In chickens the male is homogametic, while the female is heterogametic.

- (a) A male chicken with black, non-barred feathers was crossed with a female chicken with splashed-white, barred feathers. All the offspring had blue feathers, but the males were barred and the females were non-barred.
- (i) Using the symbols given on the previous page draw a genetic diagram to show this cross. [4]

Parental phenotype	Male, black, non-barred feathers	Female, splashed white, barred feathers
Genotype	$C^B C^B X^a X^a$	$C^w C^w X^A Y$
Gametes	$C^B X^a$	$C^w X^A$ $C^w Y$
Offspring genotypes	$C^B C^w X^A X^a$	$C^B C^w X^a Y$
Phenotypes	Male blue barred	Female blue non-barred

- (ii) Explain how a farmer could use a breeding programme to find out the genotype of a male chicken with blue, barred feathers. [3]
- Male chicken genotype is homozygous ($C^B C^w X^A X^A$) or heterozygous ($C^B C^w X^A X^a$) for barred feathers
 - Test Cross male chicken with a non barred female ($C^B C^w X^a Y$) and check offsprings
 - If all offsprings are barred indicates male parent is homozygous for barr, if ratio is otherwise then the male parent is heterozygous

- (b) The table below shows the genotypes and wool length in two breeds of sheep, X and Y, and their F1 hybrids.

	Genotype	Mean wool length/cm
Breed X	DDEEff	18
Breed Y	ddeeFF	14
F1 (Breed X x Breed Y)	DdEeFf	16

When the F1 were crossed among them, the F2 offspring had a mean wool length ranging from 10 cm to 22 cm.

Assume that the inheritance of wool length in sheep depends upon alleles at three loci acting additively and that all variation in wool length in the F2 population is due to the segregation of alleles at these three loci.

What proportion of the F2 offspring are expected to have a mean wool length of 22 cm? Explain your reasoning. [4]

- F2 offspring that have mean wool length of 22 cm have genotype of DDEEFF
- Genes D, E and F contribute to wool length in the same manner additively
- Probability of being homozygous at each of the locus D, E and F is $1/4$
- Probability of being DDEEFF (22 cm) = $1/4 \times 1/4 \times 1/4 = 1/64$ if three pairs of alleles segregate independently.

[Total: 11 marks]

7. **Figure 7.1** below shows the movement of O_2 and CO_2 between two organelles, A and B, in a mesophyll cell of a leaf, in the presence of light

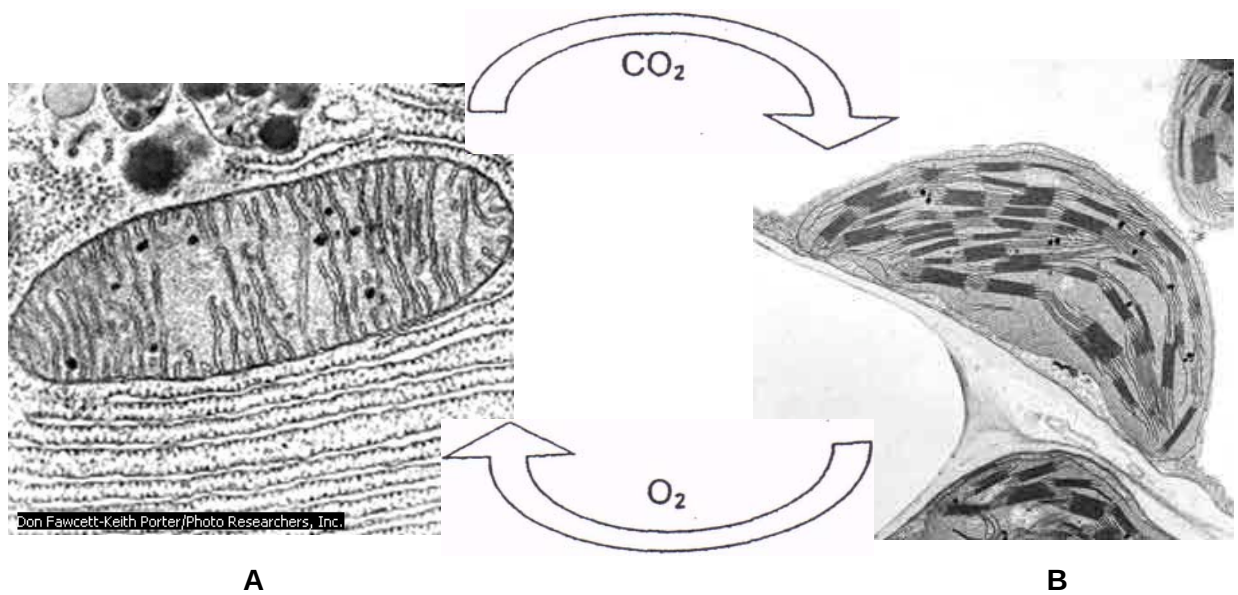


Fig 7.1 Movement of O_2 and CO_2 between organelles A and B

(a) State the process taking place in:

(i) organelle A:

- Respiration/ Krebs cycle/ Oxidative Phosphorylation

(ii) organelle B: [1]

- Photosynthesis/ Calvin cycle/ Photophosphorylation

(b) Briefly describe how organelle B is structurally adapted to carry out its function(s). [2]

- Surface area of Thylakoid/grana increase for more chlorophyll pigments/protein complex for absorbing light (any description of light harvesting)
- Membrane impermeable to H^+ (description of chemiosmosis)
- Stroma contains enzyme/ DNA and ribosomes for series of enzyme catalysed reactions in Calvin Cycle.
- Different photosystem pigments (PSI/ PS II), trap different wavelength of light

(c) ATP synthesis takes place in both organelles A and B. State three ways in which the process of ATP synthesis differs in the two organelles. [3]

- Mitochondria – electrons/protons from oxidation of organic compounds vs chloroplast from photolysis of water/chlorophylla a
- NAD/FAD vs NADP as electron carrier
- H^+ released combine with oxygen to form water vs H^+ recycled
- Oxygen required as final electron/hydrogen acceptor vs oxygen released by photolysis
- Electron from ETC (mito) combine with oxygen/ hydrogen ions to form water, electrons from ETC returned to chlorophyll molecules/ NADP

(d) Briefly describe the reaction/s that bring/s about the release of CO_2 in organelle A. [3]

- Oxidative decarboxylation
- Pyruvate converted to acetyl CoA by decarboxylase
- Citric acid (6C) converted to α - ketoglutarate (or 5-C intermediate)
- 5 carbon intermediate to be 4 carbon intermediate

(e) In humans, certain tissues can undergo anaerobic respiration if conditions make it necessary.

(i) Explain why large stores of glycogen enable these tissues to tolerate anaerobic conditions for longer period of time. [3]

- Low energy yield/ formation of only 2 ATP nett per glucose molecule
- Requires large amount of glucose for anaerobic respiration
- Glycogen broken down to glucose in these tissues

(ii) During anaerobic respiration, pyruvate is converted to lactate. Explain the significance of this conversion. [3]

- Reduced NAD produced by glycolysis must be oxidized back to NAD
- Glycolysis to continue (for more ATP) or breakdown
- Substrate – level phosphorylation for synthesis of more ATP

[Total: 15 marks]

Section B

Answer **either** question 8 **or** 9 in this section.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

8. (a) Explain how the structure of two named cellular organelles are adapted for genetic expression. [8]

- Definition of genetic expression – transcription + translation (post transcriptional processing and post translation modifications are optional)

[max 4 marks for each organelle, 1 mark for naming]

- Ribosome (free or bound to RER)
- rRNA complexed with ribosomal proteins
- form specific sites (A,P and E sites) for binding of aminoacyl-tRNAs, peptidyl-tRNAs
- able to bind to mRNA template
- ribosomal complex able to bind to various initiation, elongation and termination factors
- 3-D structure of ribosome – facilitates binding and release of various component e.g. release of polypeptide chain after translation is complete.
- Small ribosomal subunit recognizes 5' cap of mRNA
- Ribosome complex able to interact with translation initiation factors
- Large ribosome subunit has ribozyme/ catalytic/ peptidyl transferase activity for formation of peptide bonds
- Nucleus in eukaryotes [Note: applicable to eukaryotes only, failure to mention eukaryote, max 2 marks for section]
- Nuclear pores with proteins controlling movement of RNA and proteins between nucleus and cytoplasm
- Compartmentalisation of nuclear material from cytoplasm
- Allowing post transcriptional level of control and modification of transcripts before translation.
- Contain chromatin modification enzymes e.g. histone acetylases that chemically modify chromatin
- Endoplasmic reticulum
- Ribosomes stud cytosolic surface
- Having orientation such that polypeptide chain enters lumen (as translation proceeds) for post translational folding
- Lumen separate compartment from cytosol so as to maintain optimal pH and other conditions for proper folding of polypeptides
- Golgi apparatus
- System of interconnected tubules, lumen at optimal conditions for enzyme-mediated modification reactions
- Cis face receives vesicles budding off from rER
- Chemical modifications important in targeting of proteins to specific cellular locations
- Membrane vesicles can pinch off from trans face of Golgi apparatus to form secretory vesicles for exocytosis.

(b) Explain how the changes at the genome level can lead cancer. [7]

Reference to:

- Mutations in cancer critical genes
- Increasing rate of division with respect to cell death leading to uncontrolled or excessive cell division
- Cancer is a multi-step process and requires multiple/ accumulation of mutations

Proto-oncogenes [max 4]

- Gain-of-function mutations – one copy of gene mutated, dominant in nature leading to formation of oncogene
- Proto-oncogenes with the normal function in cell division, mutation to oncogene leads to uncontrolled cell proliferation
- Mutations in promoters or control elements leading to up-regulation of proto-oncogene expression
- Mutations with proto-oncogene coding region result in active gene product or gene product that binds to its substrate with increased affinity (or degradation-resistant gene product)
- AVP (Possible types of mutations that can occur: gene amplification/ translocation/ duplication/ slippage during DNA replication/ unequal crossing over/ point mutations)
- AVP (named examples)

Tumour suppressor genes [max 4]

- Loss of function mutations – both copies of the gene need to be mutated OR recessive manner.
- Tumour suppressor genes normally function in preventing inappropriate cell cycle progression, loss of function leads to inappropriate cell cycle progression.
- Mutations in promoters or control elements lead to down regulation of tumour suppressor gene expression.
- Mutations within tumour suppressor gene coding region result in inactive gene product or gene product that binds to its substrate with decreased affinity or a non-functional gene product due to loss of protein structure.
- AVP (named examples)

(c) Discuss, in terms of prokaryote and eukaryote genomes, the structure and organization of the mitochondria genome. [5]

- mtDNA similar to prokaryotic DNA;
- single circular DNA molecule, not paired or linear like eukaryotic chromosomes;
- similar to prok without nuclear membrane, found in matrix of mitochondria;
- no histone proteins, similar to prok;
- little or no non-coding DNA or introns;
- non-universal genetic code. Does not follow universal translation rules;
- some bases appear to be in two genes at once/ end of one gene is start of the next;
- contains genes coding for its own rRNA and tRNA;
- semi-autonomous reproduction;
- still requires some proteins syn from nuclear genome;
- evidence suggest origin of mtDNA from prok sources;
- ref to endosymbiotic theory;
- maternally inherited;

9. (a) Explain Neo-Darwinism and compare/contrast Neo-Darwinism with the Neutral Theory. [11]

[Max 7 for explanation of Neo-Darwinism]

- Neo-Darwinism is the idea linking genetics to Darwinism or Darwin's theory on Natural Selection
- Darwinism/Natural Selection described organism having the potential of over-reproducing/large number of offsprings, limitation of resource leading to competition.
- Individuals vary from one another/ variation exists in the population, individuals with advantageous traits/ features survive and pass advantageous traits to offspring.
- Neo-Darwinism explains that the traits are controlled by genes
- Inheritance via alleles passed down to offspring in gametes

[Max 3 for Variation caused by genes]

- Variation due to meiosis/ gamete formation
- Crossing over and independent assortment
- Random fertilization
- Epigenetics
- Transposons
- Ultimate source of variation from mutation
- Chromosomal and gene mutation

[Max 4 for compare/contrast Neo-Darwinism with Neutral Theory]

- Similarities - Advantages mutations are favoured and selected while disadvantages traits are removed during selection.
- Differences – Frequency of appearance of genes, Neo-Darwinism (Selection theory) depends more on selection than genetic drift (Neutral Theory)
- Differences – Evolution of population due mainly to neutral alleles vs due to the selection of advantageous alleles.
- Differences – Mutations defined in Good, Neutral and Bad (Neutral Theory) compared to Good/Bad mutations (Neo-Darwinism)

(b) Discuss the advantages of molecular methods in tracing of human ancestry using mitochondrial DNA. [5]

- maternally inherited;
- sperm mitochondria lost at fertilization;
- possibly to track maternal ancestry
- higher mutation rate c.f with nuclear genome;
- constant mutation rate allows mtDNA to be used as molecular clock;
- in any given stretch of DNA, it mutates at a reliable rate which can be estimated/ dated. Divergence is estimated by calculate the rate at which a stretch of DNA changes;
- better than nuclear DNA because it can be used to follow ancestry over very long time periods;
- nuclear genome involves fusion of gametes. Not possible to differentiate between paternal and maternal source of DNA. Mitochondria DNA does not have mixture of paternal & maternal DNA;
- gametes formed may be recombinant because of crossing over/ random segregation during meiosis. MtDNA does not have mixture;
- when 2 mtDNA sequences are compared, differences show mutations between one generation;

(c) Suggest reasons why Human is believed to have evolved from apes. [4]

[Note – The question is hypothetical and suggestion of possible links are expected]

- Comparative anatomy – based on a common prototype/ tracking through fossil chronology
- Homologous structures
- Vestigial structures – reduced tail bones/ gripping power
- Molecular homology – comparison between similarities of DNA base sequence.
- AVP